Formulae:

1. If $\mathcal{L}[f(t)] = F(s)$, then $\mathcal{L}[t^n f(t)] = (-1)^n \frac{d^n}{ds^n}[F(s)]$.
2. If $\mathcal{L}[f(t)] = F(s)$, then $\mathcal{L}\left[\frac{f(t)}{t}\right] = \int_s^\infty F(s)\,ds$.
3. If $\mathcal{L}[f(t)] = F(s)$, then $\mathcal{L}\left[\int_0^t f(t)\,dt\right] = \frac{1}{s}F(s)$.

problem:

Find the laplace transform of:

(a) $t\cos at$ (b) $t\sin at$ (c) $te^t\sin 4t$ (d) $f(t) = \int_0^t \frac{\sin t}{t}\,dt$.

Solution:

(a) $t\cos at$

Let $f(t) = \cos at$

$\therefore \mathcal{L}(\cos at) = \frac{s}{s^2+a^2}$

$$\therefore \mathcal{L}(t\cos at) = (-1)^1 \frac{d}{ds}\left[\frac{s}{s^2+a^2}\right]$$

$$= -\frac{d}{ds}\left[\frac{s}{s^2+a^2}\right]$$

$$= -\left[\frac{(s^2+a^2)\frac{d}{ds}(s) - s\frac{d}{ds}(s^2+a^2)}{(s^2+a^2)^2}\right]$$

$$= -\left[\frac{(s^2+a^2)\cdot 1 - s(2s+0)}{(s^2+a^2)^2}\right]$$

$$= -\left[\frac{s^2+a^2-2s^2}{(s^2+a^2)^2}\right] = -\left[\frac{a^2-s^2}{(s^2+a^2)^2}\right]$$

$$\therefore \mathcal{L}(t\cos at) = \frac{s^2-a^2}{(s^2+a^2)^2} \text{ (Ans:)}$$

(b) $t \sin at$

Let $f(t) = \sin at$

$\therefore \mathcal{L}(\sin at) = \frac{a}{s^2+a^2}$

$\therefore \mathcal{L}(t \sin at) = (-1)^1 \frac{d}{ds}\left[\frac{a}{s^2+a^2}\right]$

$= -\frac{d}{ds}\left[\frac{a}{s^2+a^2}\right]$

$= -\left[\frac{(s^2+a^2)\frac{d}{ds}(a) - a\frac{d}{ds}(s^2+a^2)}{(s^2+a^2)^2}\right]$

$= -\left[\frac{(s^2+a^2)\cdot 0 - a(2s+0)}{(s^2+a^2)^2}\right]$

$= -\left[\frac{0-2as}{(s^2+a^2)^2}\right]$

$\therefore \mathcal{L}(t\sin at) = \frac{2as}{(s^2+a^2)^2}$ (Ans:)

(c) $te^t \sin 4t$

Let $f(t) = \sin 4t$

$\therefore \mathcal{L}(\sin 4t) = \frac{4}{s^2+4^2} = \frac{4}{s^2+16}$ $\left[\because \mathcal{L}(\sin at) = \frac{a}{s^2+a^2}\right.$

$\therefore \mathcal{L}(e^t \sin 4t) = \frac{4}{(s-1)^2+4^2} = \frac{4}{(s-1)^2+16}$ $\left[\because \mathcal{L}(e^{at}\sin bt) = \frac{b}{(s-a)^2+b^2}\right.$

$= \frac{4}{s^2-2s+1+16}$

$\Rightarrow \mathcal{L}(e^t \sin 4t) = \frac{4}{s^2-2s+17}$

$$\therefore \mathcal{L}\left(t e^{t}\sin 4t\right) = (-1)^{1}\frac{d}{ds}\left[\frac{1}{s^2-2s+17}\right]$$

$$= -\frac{d}{ds}\left[\frac{1}{s^2-2s+17}\right]$$

$$= -\left[\frac{(s^2-2s+17)\frac{d}{ds}(1) - 1\frac{d}{ds}(s^2-2s+17)}{(s^2-2s+17)^2}\right]$$

$$= -\left[\frac{(s^2-2s+17)\cdot 0 - 1(2s-2\cdot 1+0)}{(s^2-2s+17)^2}\right]$$

$$= -\left[\frac{0-8s+8}{(s^2-2s+17)^2}\right]$$

$$= \frac{8s-8}{(s^2-2s+17)^2}$$

$$= \frac{8(s-1)}{(s^2-2s+17)^2} \text{ (Ans:)}$$

Inverse laplace transform:

If $\mathcal{L}[f(t)] = F(s)$, then $\mathcal{L}^{-1}[F(s)] = f(t)$

where $\mathcal{L}^{-1}$ is called the inverse Laplace transform operator.

(d) $f(t) = \int_0^t \frac{\sin t}{t} dt$

$\because \mathcal{L}(\sin t) = \frac{1}{s^2+1^2} = \frac{1}{s^2+1}$ $\left[\because \mathcal{L}(\sin at) = \frac{a}{s^2+a^2}\right.$

$\because \mathcal{L}\left(\frac{\sin t}{t}\right) = \int_s^\infty \frac{1}{s^2+1} ds$ $\left[\because L\left[\frac{f(t)}{t}\right] = \int_s^\infty F(s) ds\right.$

$= \int_s^\infty \frac{ds}{1+s^2}$

$= \left[\tan^{-1} s\right]_s^\infty$

$= \tan^{-1}\infty - \tan^{-1} s$

$= \tan^{-1}\tan\frac{\pi}{2} - \tan^{-1} s$

$= \frac{\pi}{2} - \tan^{-1} s$

$\because \mathcal{L}\left[\int_0^t \frac{\sin t}{t} dt\right] = \frac{1}{s}\left[\frac{\pi}{2} - \tan^{-1} s\right]$ (Ans:)

$\left[\because \mathcal{L}\left[\int_0^t \frac{f(t)}{t} dt\right] = \frac{1}{s} F(s).\right.$

Important formulae:

1. $\mathcal{L}^{-1}\left(\frac{1}{s}\right)=1$
2. $\mathcal{L}^{-1}\left(\frac{1}{s-a}\right)=e^{at}$
3. $\mathcal{L}^{-1}\left(\frac{a}{s^2+a^2}\right)=\sin at$
4. $\mathcal{L}^{-1}\left(\frac{s}{s^2+a^2}\right)=\cos at$
5. $\mathcal{L}^{-1}\left[\frac{b}{(s-a)^2+b^2}\right]=e^{at}\sin bt$
6. $\mathcal{L}^{-1}\left[\frac{s-a}{(s-a)^2+b^2}\right]=e^{at}\cos bt$

Example of different formulae:

Formula #02: $\mathcal{L}^{-1}\left(\frac{1}{s-a}\right)=e^{at}$

(a) $\mathcal{L}^{-1}\left(\frac{1}{s-2}\right)=e^{2t}$

(b) $\mathcal{L}^{-1}\left(\frac{1}{s+3}\right)=\mathcal{L}^{-1}\left[\frac{1}{s-(-3)}\right]=e^{-3t}$

(c) $\mathcal{L}^{-1}\left(\frac{1}{s-\frac{1}{2}}\right)=e^{\frac{t}{2}}$

(d) $\mathcal{L}^{-1}\left(\frac{1}{s+\frac{1}{3}}\right)=\mathcal{L}^{-1}\left[\frac{1}{s-(-\frac{1}{3})}\right]=e^{-\frac{t}{3}}$

Formula #03: $\mathcal{L}^{-1}\left(\frac{a}{s^2+a^2}\right) = \sin at$

(a) $\mathcal{L}^{-1}\left(\frac{2}{s^2+4}\right) = \mathcal{L}^{-1}\left(\frac{2}{s^2+2^2}\right) = \sin 2t$

(b) $\mathcal{L}^{-1}\left(\frac{\sqrt{2}}{s^2+2}\right) = \mathcal{L}^{-1}\left(\frac{\sqrt{2}}{s^2+(\sqrt{2})^2}\right) = \sin \sqrt{2}\,t$

(c) $\mathcal{L}^{-1}\left(\frac{\frac{1}{9}}{s^2+\frac{1}{81}}\right) = \mathcal{L}^{-1}\left[\frac{\frac{1}{9}}{s^2+\left(\frac{1}{9}\right)^2}\right] = \sin \frac{t}{9}$

Formula #04: $\mathcal{L}^{-1}\left(\frac{s}{s^2+a^2}\right) = \cos at$

(a) $\mathcal{L}^{-1}\left(\frac{s}{s^2+9}\right) = \mathcal{L}^{-1}\left(\frac{s}{s^2+3^2}\right) = \cos 3t$

(b) $\mathcal{L}^{-1}\left(\frac{s}{s^2+7}\right) = \mathcal{L}^{-1}\left(\frac{s}{s^2+(\sqrt{7})^2}\right) = \cos \sqrt{7}\,t$

(c) $\mathcal{L}^{-1}\left(\frac{s}{s^2+\frac{1}{2}}\right) = \mathcal{L}^{-1}\left[\frac{s}{s^2+\left(\frac{1}{\sqrt{2}}\right)^2}\right] = \cos \frac{t}{\sqrt{2}}$.

Formula #05: $\mathcal{L}^{-1}\left[\frac{b}{(s-a)^2+b^2}\right] = e^{at}\sin bt$

(a) $\mathcal{L}^{-1}\left[\frac{3}{(s-2)^2+9}\right] = \mathcal{L}^{-1}\left[\frac{3}{(s-2)^2+3^2}\right] = e^{2t}\sin 3t$

(b) $\mathcal{L}^{-1}\left[\frac{\sqrt{2}}{(s+3)^2+2}\right] = \mathcal{L}^{-1}\left[\frac{\sqrt{2}}{\{s-(-3)\}^2+(\sqrt{2})^2}\right] = e^{-3t}\sin \sqrt{2}\,t$

(c) $\mathcal{L}^{-1}\left[\frac{4}{(s-\frac{1}{2})^2+16}\right] = \mathcal{L}^{-1}\left[\frac{4}{(s-\frac{1}{2})^2+4^2}\right] = e^{\frac{t}{2}}\sin 4t$

Formula #06: $\mathcal{L}^{-1}\left[\frac{s-a}{(s-a)^2+b^2}\right]=e^{at}\cos bt$

(a) $\mathcal{L}^{-1}\left[\frac{s-3}{(s-3)^2+4}\right]=\mathcal{L}^{-1}\left[\frac{s-3}{(s-3)^2+2^2}\right]=e^{3t}\cos 2t$

(b) $\mathcal{L}^{-1}\left[\frac{s+2}{(s+2)^2+9}\right]=\mathcal{L}^{-1}\left[\frac{s-(-2)}{\{s-(-2)\}^2+3^2}\right]=e^{-2t}\cos 3t$

(c) $\mathcal{L}^{-1}\left[\frac{s-\frac{1}{2}}{(s-\frac{1}{2})^2+7}\right]=\mathcal{L}^{-1}\left[\frac{s-\frac{1}{2}}{(s-\frac{1}{2})^2+(\sqrt{7})^2}\right]=e^{\frac{t}{2}}\cos\sqrt{7}t.$

problem:

Find the inverse laplace transform of:

(a) $\frac{1}{s-2}$ (b) $\frac{1}{s+3}$ (c) $\frac{1}{s-\frac{1}{2}}$ (d) $\frac{1}{s+\frac{1}{2}}$ (e) $\frac{1}{2s-7}$

(f) $\frac{1}{3s+5}$ (g) $\frac{3}{s^2+9}$ (h) $\frac{1}{s^2+16}$ (i) $\frac{1}{s^2+7}$ (j) $\frac{s}{s^2+16}$

(k) $\frac{s}{s^2+3}$ (l) $\frac{3}{s^2-4s+13}$ (m) $\frac{\sqrt{2}}{s^2-6s+11}$

(n) $\frac{4}{s^2-2s+17}$ (o) $\frac{s-2}{s^2-2s+13}$ (p) $\frac{s-3}{s^2-6s+14}$

Solution:

(a) $\frac{1}{s-2}$

Let $F(s)=\frac{1}{s-2}$

$\therefore \mathcal{L}^{-1}\left[\frac{1}{s-2}\right]=e^{2t}$ (Ans:) $\left[\because \mathcal{L}^{-1}\left(\frac{1}{s-a}\right)=e^{at}\right.$

(b) $\frac{1}{s+3}$

Let $F(s)=\frac{1}{s+3}$

$\therefore \mathcal{L}^{-1}\left(\frac{1}{s+3}\right)=\mathcal{L}^{-1}\left[\frac{1}{s-(-3)}\right]=e^{-3t}$ (Ans:)

(e) $\frac{1}{2s-7}$

Let $F(s)=\frac{1}{2s-7}$

$\therefore \mathcal{L}^{-1}\left(\frac{1}{2s-7}\right)=\mathcal{L}^{-1}\left[\frac{1}{2(s-\frac{7}{2})}\right]=\frac{1}{2}\mathcal{L}^{-1}\left(\frac{1}{s-\frac{7}{2}}\right)=\frac{1}{2}e^{\frac{7}{2}t}$

(g) $\frac{3}{s^2+9}$

Let $F(s)=\frac{3}{s^2+9}$

$\therefore \mathcal{L}^{-1}\left(\frac{3}{s^2+9}\right)=\mathcal{L}^{-1}\left(\frac{3}{s^2+3^2}\right)=\sin 3t$ $\left[\because \mathcal{L}^{-1}\left(\frac{a}{s^2+a^2}\right)=\sin at\right.$

(h) $\frac{1}{s^2+16}$

Let $F(s)=\frac{1}{s^2+16}$

$\therefore \mathcal{L}^{-1}\left(\frac{1}{s^2+16}\right)=\mathcal{L}^{-1}\left(\frac{1}{s^2+4^2}\right)=\mathcal{L}^{-1}\left(\frac{\frac{1}{4}\cdot 4}{s^2+4^2}\right)$

$=\frac{1}{4}\mathcal{L}^{-1}\left(\frac{4}{s^2+4^2}\right)$

$=\frac{1}{4}\sin 4t$ (Ans:)

(j) $\frac{s}{s^2+16}$

Let $F(s) = \frac{s}{s^2+16}$

$\therefore \mathcal{L}^{-1}\left(\frac{s}{s^2+16}\right) = \mathcal{L}^{-1}\left(\frac{s}{s^2+4^2}\right) = \cos 4t$

$\left[\because \mathcal{L}^{-1}\left(\frac{s}{s^2+a^2}\right) = \cos at\right.$

(K) $\frac{s}{s^2+3}$

Let $F(s) = \frac{s}{s^2+3}$

$\therefore \mathcal{L}^{-1}\left(\frac{s}{s^2+3}\right) = \mathcal{L}^{-1}\left(\frac{s}{s^2+(\sqrt{3})^2}\right) = \cos\sqrt{3}t$ (Ans:)

(l) $\frac{3}{s^2-4s+13}$

Let $F(s) = \frac{3}{s^2-4s+13}$

$\therefore \mathcal{L}^{-1}\left(\frac{3}{s^2-4s+13}\right) = \mathcal{L}^{-1}\left(\frac{3}{s^2-4s+4+9}\right)$

$= \mathcal{L}^{-1}\left[\frac{3}{(s-2)^2+3^2}\right]$

$= e^{2t}\sin 3t$ $\left[\because \mathcal{L}^{-1}\left[\frac{b}{(s-a)^2+b^2}\right] = e^{at}\sin bt\right.$

(m) $\frac{\sqrt{2}}{s^2-6s+11}$

Let $F(s)=\frac{\sqrt{2}}{s^2-6s+11}$

$$\therefore \mathcal{L}^{-1}\left(\frac{\sqrt{2}}{s^2-6s+11}\right)=\mathcal{L}^{-1}\left(\frac{\sqrt{2}}{s^2-6s+9+2}\right)$$

$$=\mathcal{L}^{-1}\left[\frac{\sqrt{2}}{(s-3)^2+(\sqrt{2})^2}\right]$$

$$=e^{3t}\sin\sqrt{2}\,t \text{ (Ans.)}$$

(P) $\frac{s-3}{s^2-6s+14}$

Let $F(s)=\frac{s-3}{s^2-6s+14}$

$$\therefore \mathcal{L}^{-1}\left(\frac{s-3}{s^2-6s+14}\right)=\mathcal{L}^{-1}\left(\frac{s-3}{s^2-6s+9+5}\right)$$

$$=\mathcal{L}^{-1}\left[\frac{s-3}{(s-3)^2+(\sqrt{5})^2}\right]$$

$$=e^{3t}\cos\sqrt{5}t \quad \left[\because \mathcal{L}^{-1}\left[\frac{s-a}{(s-a)^2+b^2}\right]=e^{at}\cos bt\right]$$

Application of laplace transform.